

Invitation to invariant quasimorphisms

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j/w

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Setting

$$\begin{cases} G: \text{a group} \\ N \trianglelefteq G \text{ (normal subgroup)} \end{cases}$$

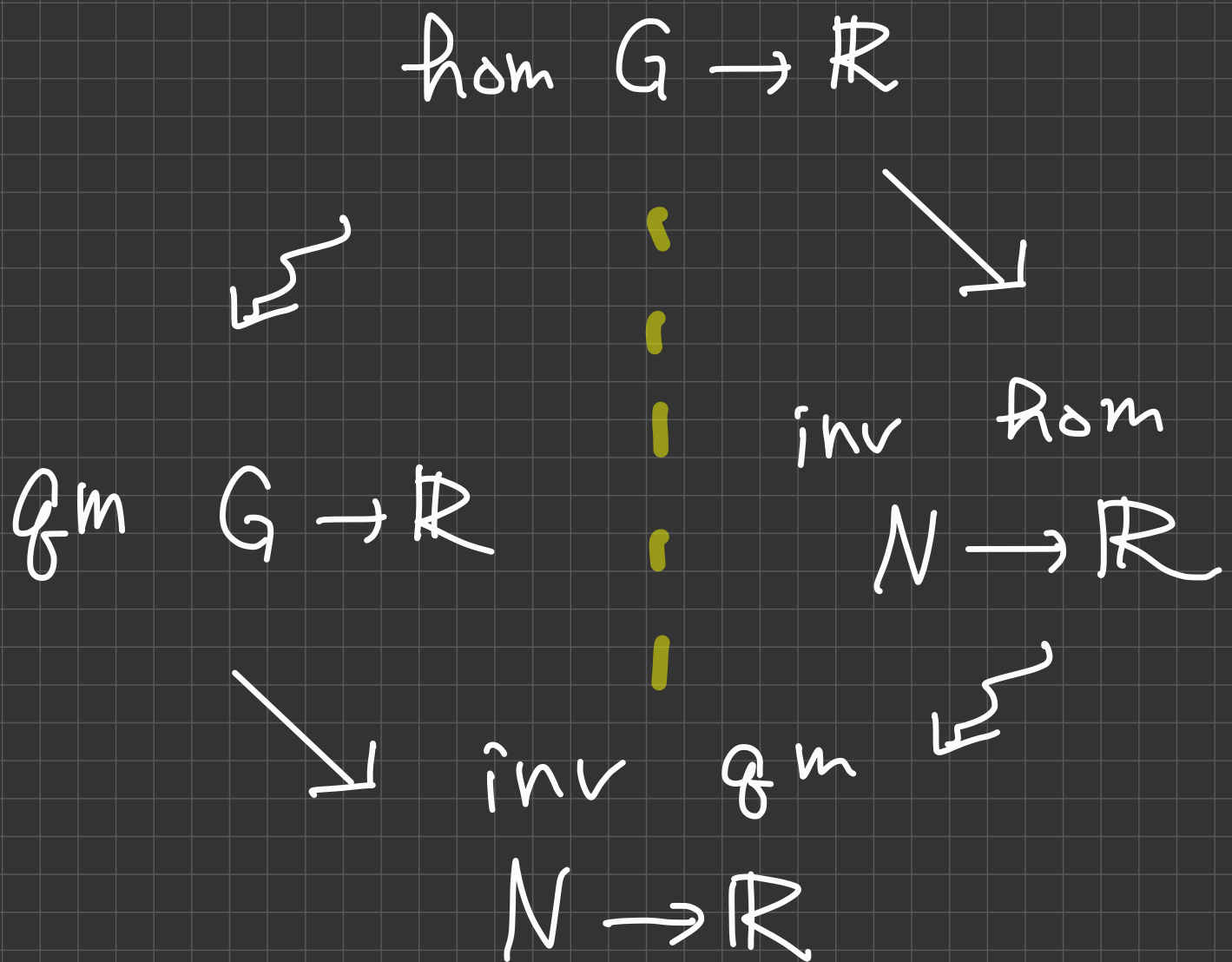
equivalently,

$$1 \rightarrow N \xrightarrow{i} G \xrightarrow{p} \Gamma \rightarrow 1$$

$\parallel$
 G/N

Short exact sequence
of groups

Overview



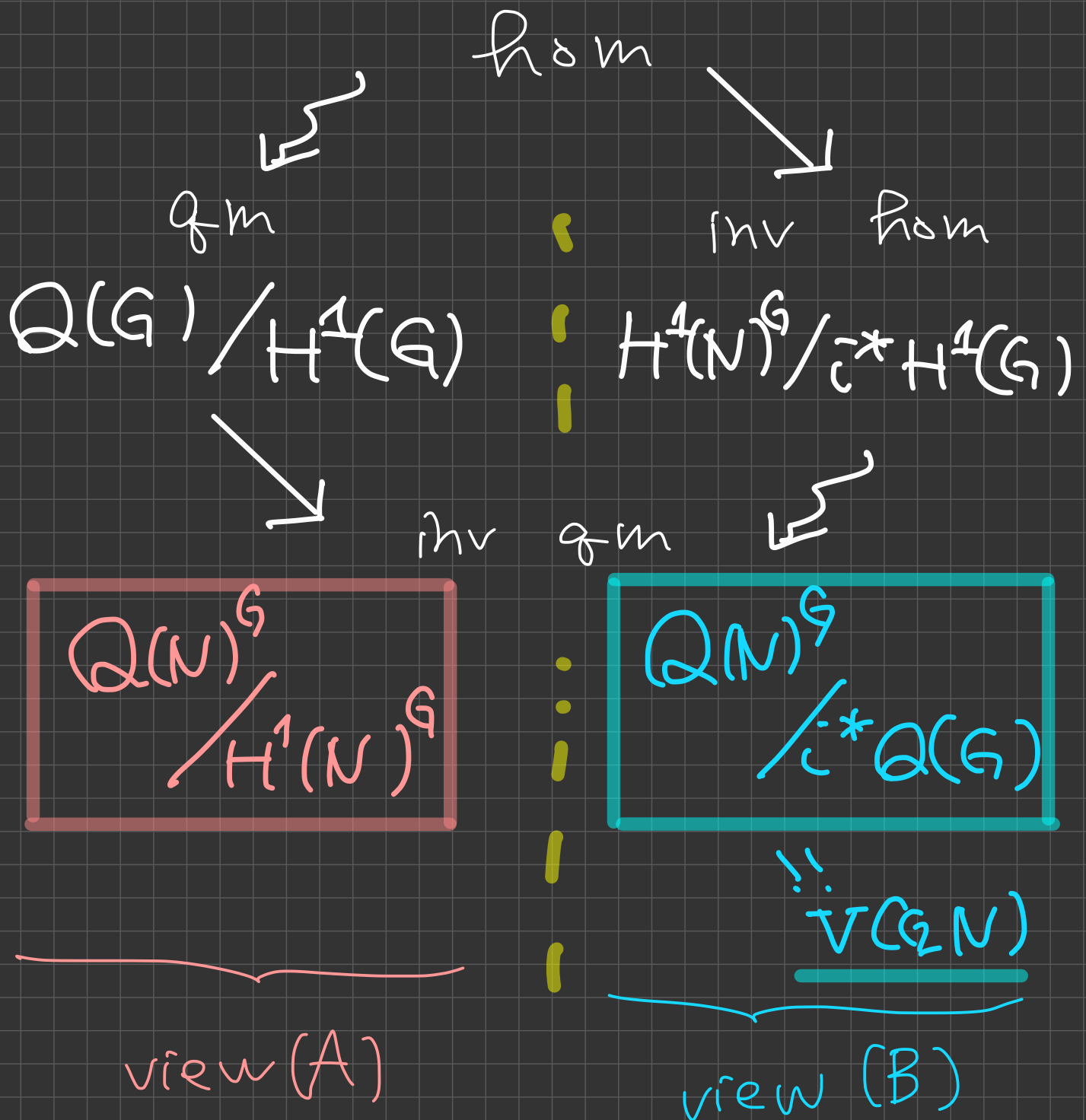
view (A)

view (B)

$\rightsquigarrow$: quasification

$\rightarrow$: relativization

Vector spaces



$$W(G, N) := Q(N)^G / H^1(N)^G + i^*Q(G)$$

: the common coterminal.

Applications

view(A) : Coarse geometry of
scl & mixed scl

e.g. $G = \pi_1(\Sigma_l)$, $l \geq 2$ 

$$N = [G, G]$$

$$(1 \rightarrow N \rightarrow G \rightarrow \Gamma = H_1(\Sigma_l; \mathbb{Z}) \rightarrow 1)$$

view(B) : Symplectic geometry
flux hom & bilinear form
on the first cohomology

e.g. $G (= \widehat{\text{Symp}}_0(\Sigma_l, \omega)) = \text{Symp}_0(\Sigma_l, \omega)$

$(l \geq 2)$ $N (= \widehat{\text{Ham}}(\Sigma_l, \omega)) = \text{Ham}(\Sigma_l, \omega)$

$$(1 \rightarrow N \rightarrow G \xrightarrow{\text{Flux}} \Gamma = H^1(\Sigma_l; \mathbb{R}) \rightarrow 1)$$

Fact (BANYAGA)

$$N = [N, N] \quad (\text{so } N = [G, G])$$

1. $V \not\subseteq W$

$$H^1(G) = H^1(G; \mathbb{R})$$

$\simeq$

$$\underline{Q(G)} := \{ \phi: G \rightarrow \mathbb{R} \mid \left. \begin{array}{l} \text{homogeneous} \\ q_m \end{array} \right\}$$

$$\cdot) \quad q_m \stackrel{\text{def}}{\iff} \underline{D(\phi)} := \sup_{f, g \in G} | \phi(fg) - \phi(f) - \phi(g) | < \infty$$

defect

$$\cdot) \quad \text{homogeneous} \stackrel{\text{def}}{\iff} \forall g \in G; \forall n \in \mathbb{Z}; \quad \phi(g^n) = n \phi(g)$$

Lem $\forall \hat{\phi}: G \rightarrow \mathbb{R} \text{ } q_m; \exists (?) \underline{\phi} \in Q(G);$

$\| \hat{\phi} - \phi \|_{\infty} < \infty$

$$\phi(g) = \lim_{k \rightarrow \infty} \frac{\hat{\phi}(g^k)}{k} \quad \text{homogenization}$$

$$(\hat{\phi} = (\text{good part}) + (\text{bounded noise}))$$

$N \trianglelefteq G$ $\angle: N \rightarrow \mathbb{R}$ is G -inv

$$\stackrel{\text{def}}{\iff} \forall g \in G; \forall x \in N; \angle(gxg^{-1}) = \angle(x)$$

$\leadsto$ can define $Q(N)^G$ $\supseteq$ $H^1(N)^G$

$$\text{Lem } Q(G) = Q(G)^G$$

$\odot (\subseteq)$

ϕ

$f, g \in G \quad k \in \mathbb{N}$

Then

$$k \phi(f g f^{-1}) = \phi((f g f^{-1})^k) \\ = \phi(f g^k f^{-1})$$

$$\left(\begin{array}{l} a \sim_D b \\ \Leftrightarrow |a-b| \leq D \end{array} \right)$$

$$\sim_{D(\phi)} \phi(f) + \phi(g^k) + \phi(f^{-1})$$

$$= k \phi(g)$$



So, $i: N \hookrightarrow G$ induces

$$i^*: Q(G) \rightarrow Q(N)^G; \phi \mapsto i^* \phi = \phi|_N \\ \cup \quad \cup \\ H^1(G) \rightarrow H^1(N)^G$$

three vector spaces

$$1) Q(N)^G / H^1(N)^G$$

$$2) Q(N)^G / i^* Q(G) =: \underline{V(G, N)}$$

$$3) Q(N)^G / H^1(N)^G + i^* Q(G) =: \underline{W(G, N)}$$

Interestingly, V & $\overline{W}$ are
finite dim'l in several cases,
 while $Q(N)^G / H^1(N)^G$ is often huge.

Thm (EPSTEIN-FUJIWARA, BESTVINA-F.
 OSIN, ...)

G : Gromov-hyperbolic, non-elementary

$N \trianglelefteq G$: infinite normal subgroup

(more generally, G : acylindrically hyp. countable)

$$\Rightarrow \dim_{\mathbb{R}} Q(N)^G / H^1(N)^G = 2^{\aleph_0}$$

Prop (K^2M^2 , 22)

If $1 \rightarrow N \rightarrow G \xrightarrow{p} \Gamma \rightarrow 1$ virtually splits,

(i.e. $\exists \Lambda \leq \Gamma$; $\exists \Delta: \Lambda \rightarrow G$ hom; finite index $p \circ \Delta = \text{id}_{\Lambda}$)

then $V = 0$

In particular, $V = 0$ if Γ virtually free

Thm (TAO, 25+)

$V = 0$ if Γ Gromov-hyperbolic

Thm 1 ($K^2 M^3$, '25)

$$1 \rightarrow N \rightarrow G \xrightarrow{p} \Gamma \rightarrow 1$$

Assume that Γ is

boundedly 3-acyclic

$$H_b^2(\Gamma) = H_b^3(\Gamma) = 0$$

= bounded cohomology

(e.g. amenable group; $SL(n, \mathbb{Z})$, $n \geq 3$)

Then,

$$W(G, N)$$

$$\cong \text{Im} \left\{ \overset{\text{comparison map}}{C_G^2}: H_b^2(G) \rightarrow H^2(G) \right\} \\ \cap \text{Im} \left\{ p^*: H^2(\Gamma) \rightarrow H^2(G) \right\}$$

In particular, in this case,

$$\dim_{\mathbb{R}} W(G, N) \leq \min \left\{ \dim_{\mathbb{R}} H^2(G), \dim_{\mathbb{R}} H^2(\Gamma) \right\}$$

e.g.

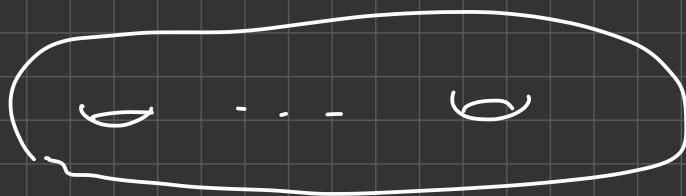
(1) F : free group, $K \trianglelefteq F$

$\Lambda = F/K$ boundedly 3-aspherical

$$\Rightarrow W(F, K) = 0$$

$$(\because H^2(F) = 0)$$

(2) $G = \pi_1(\Sigma_g)$, $g \geq 2$



$$N = [G, G]$$

$$\Rightarrow \dim_{\mathbb{R}} W(G, N) = \underline{1}$$

$$(H^2(G) \cong H^2(\Sigma_g; \mathbb{R}) \cong \mathbb{R})$$

course geometry of

2. $\text{view}(A)$: scl_G & $\text{scl}_{G,N}$

Def

$$gzg^{-1}z^{-1}$$

$$\cdot) \mathcal{L}_{G,N} := \{ [g, z] \mid \begin{matrix} g \in G \\ z \in N \end{matrix} \}$$

$$\cdot) \underline{[G, N]} := \langle \mathcal{L}_{G,N} \rangle$$

the mixed commutator subgroup

$\text{cl}_{G,N}$: word length w.r.t. $\mathcal{L}_{G,N}$

the mixed commutator length

$$\cdot) \underline{\text{scl}_{G,N}} : \text{scl}_{G,N}(g) := \lim_{k \rightarrow \infty} \frac{\text{cl}_{G,N}(g^k)}{k}$$

the mixed scl

stable commutator length

Original setting: $N = G$

$$\text{scl}_{G,G} =: \text{scl}_G$$

original: BAVARD 1991

Thm 2 ($K^2 N^2$: mixed BAVARD duality)

$$\forall y \in [G, N];$$

$$\text{scl}_{G,N}(y) = \sup_{\substack{[\mu] \in \frac{\mathbb{Q}(N)^G}{H^1(N)^G}} \frac{|\mu(y)|}{2D(\mu)}$$

$$\left(\frac{\mathbb{Q}(N)^G}{H^1(N)^G}, 2D \right) \times ([G, N], \text{scl}_{G,N}) \rightarrow \mathbb{R}$$
$$\downarrow \qquad \qquad \qquad \downarrow$$
$$([\mu], y) \longmapsto \mu(y)$$

Coarse geometry of scl

motivation:

$$\forall g_1, g_2 \in [G, N]:$$

$$\text{scl}_{G,N}(g_1 g_2) \leq \text{scl}_{G,N}(g_1) + \text{scl}_{G,N}(g_2) + \frac{1}{2}$$

}

by Mixed Bavard
(unavoidable !!)

$$d\text{scl}_{G,N}^+ : N \times N \rightarrow \mathbb{R} \cup \{\infty\}$$

$$d\text{scl}_{G,N}^+(x_1, x_2) := \begin{cases} \text{scl}_{G,N}(x_1^{-1}x_2) + \frac{1}{2} & \text{if } x_1 \neq x_2 \\ 0 & \text{if } x_1 = x_2 \end{cases}$$

is $\left. \begin{array}{l} \text{bi-}N\text{-inv} \\ G\text{-conj-inv} \end{array} \right\}$ metric on N

Coarse kernel:

INANOV,

Def ($\stackrel{\text{cf}}{=}$ LEITNER-VIGOLD)

$(G, d_G), (H, d_H)$ groups
w/ bi-inv metrics.

$$\bar{\Phi} : (G, d_G) \rightarrow (H, d_H)$$

coarse hom

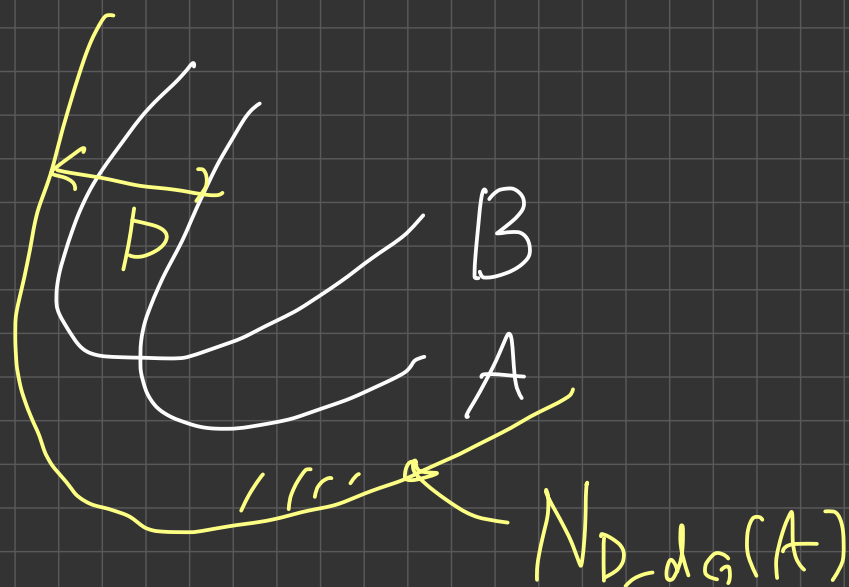
$$\left(\begin{array}{l} \exists c \geq 0 ; \bar{\Phi}(g_1, g_2) \sim \bar{\Phi}(g_1) \cdot \bar{\Phi}(g_2) \\ c \text{ in } d_H \\ \& \text{ bornologous (controlled)} \end{array} \right)$$

$A \subseteq G$ is a coarse kernel
of $\bar{\Phi}$

$\Leftrightarrow$.) $\Phi(A)$ is d_H -bdd
 $\Leftarrow$ f

.) A is 'coarsely largest'
 in d_G among such.

$\forall B \subseteq G$ with $\Phi(B)$ d_H -bdd;
 $\exists D \geq 0; N_{D, d_G}(A) \supseteq B$



Rem Coarse kernel may not exist.

(e.g. $(\ell_2, \|\cdot\|_2) \rightarrow (\ell_\infty, \|\cdot\|_\infty); \mathbb{Z} \rightarrow \mathbb{Z}$)

$$\begin{array}{ccc}
 \begin{array}{c} \uparrow \eta^y \\ ([G, N], \text{sd}_{G, N}) \end{array} & \xrightarrow{\quad} & \begin{array}{c} \uparrow \eta^y \\ ([G, G], \text{sd}_G) \end{array} \\
 \uparrow \text{"dual"} & & \uparrow \text{"dual"} \\
 Q(N)^G / H^1(N)^G & \xleftarrow{\quad \overline{\cdot}^* \quad} & Q(G) / H^1(G) \\
 \nwarrow & & \nearrow \\
 C_{\text{kernel}} = W(G, N) & &
 \end{array}$$

Thm 3 ($K^2 M^3$, 25+)

$$N = [G, G] \quad \text{with } m$$

Assume that $\dim_{\mathbb{R}} W(G, N) < \infty$

Then,

$$\left(\begin{array}{c} \text{the coarse kernel} \\ \text{of } ([G, N], \text{sd}_{G, N}) \\ \rightarrow ([G, G], \text{sd}_G) \end{array} \right) \cong (\mathbb{Z}^m, \|\cdot\|_1) \text{ as coarse groups}$$

e.g. $G = \pi_1(\Sigma_g) \quad g \geq 2$

$$= \langle a_1, \dots, a_g, b_1, \dots, b_g \mid [a_1, b_1] \dots [a_g, b_g] \rangle$$

(Recall $\dim_{\mathbb{R}} W(G, N) = 1$)

Take $\mu \in \mathbb{Q}(N)^g$ with $[\mu] \neq 0$ in $W(G, N)$.

$$F := \langle \alpha_1, \dots, \alpha_g, \beta_1, \dots, \beta_g \mid - \rangle$$

free group

$$\pi: F \twoheadrightarrow G; \quad \begin{array}{l} \alpha_i \mapsto a_i \\ \beta_i \mapsto b_i \end{array}$$

$$K := [F, F]$$

$$j: K \hookrightarrow F$$

$$\begin{array}{c} \pi^* \mu \in \mathbb{Q}(K)^F \\ \uparrow \\ \mu \in \mathbb{Q}(N)^g \end{array} \quad \begin{array}{c} \uparrow \pi^* \\ \mu \in \mathbb{Q}(N)^g \end{array}$$

$$\left(\begin{array}{c} F \supseteq K = [F, F] \\ \downarrow \pi \\ G \supseteq N = [G, G] \end{array} \right)$$

key $W(F, K) = 0 \iff F: \text{free}$
 $F/K: \text{bdd 3-acyc.}$
 (i.e. $Q(F)^K = H^1(\bar{F})^K + j^* Q(F)$)

So, can write

$$\pi^* \mu = \underbrace{h}_{H^1(K)^F} + j^* \underbrace{\phi}_{Q(F)}$$

Note: $h^* \in H^1(K)^F$
 descends to $\bar{h}^* \in H^1(K)^G$
 $\iff h^*([\alpha_1, \beta_1] \dots [\alpha_l, \beta_l]) = 0$

Can show:

$$h([\alpha_1, \beta_1] \dots [\alpha_l, \beta_l]) \neq 0.$$

Now, consider the map $(\star)$

$$\begin{aligned} \bar{\Psi} : (\mathbb{Z}, 1.1) &\longrightarrow ([G, N] \text{ s.d.g.N}) \\ \psi &\qquad \qquad \qquad \psi \\ k &\longmapsto \pi([\alpha_1^k, \beta_1] \cdots [\alpha_l^k, \beta_l]) \\ &\qquad \qquad \qquad \parallel \\ &\qquad \qquad \qquad [\alpha_1^k, b_1] \cdots [\alpha_l^k, b_l] \end{aligned}$$

Then:

$$\cdot) \quad \forall k \in \mathbb{Z}; \quad \text{cl}_G(\bar{\Psi}(k)) \leq l$$

So, $\bar{\Psi}(\mathbb{Z})$ is s.d.g.-bounded

$$\cdot) \quad \mu(\bar{\Psi}(k))$$

$$= \pi^* \mu([\alpha_1^k, \beta_1] \cdots [\alpha_l^k, \beta_l])$$

$$= h([\alpha_1^k, \beta_1] \cdots [\alpha_l^k, \beta_l])$$

$$+ \underbrace{\phi([\alpha_1^k, \beta_1] \cdots [\alpha_l^k, \beta_l])}$$

$$\sum_0^{2l} d(\phi)$$

$$\sim h([\alpha_1^k, \beta_1] \dots [\alpha_l^k, \beta_l])$$

$2\ell(\Phi)$

$$= k \underbrace{h([\alpha_1, \beta_1] \dots [\alpha_l, \beta_l])}$$

$\uparrow$ $\neq 0$ by $(\star)$

by commutator calculus

$$\left(\begin{array}{l} \forall f_1, f_2, f_3 \in F; \\ [f_1 f_2, f_3] = (f_1 [f_2, f_3] f_1^{-1}) \\ \qquad \qquad \qquad [f_1, f_3] \end{array} \right)$$

So, mixed Bavard implies:

$$(\mathbb{F}(\mathbb{Z}), \text{sol}_{\mathbb{Z}} N) \cong (\mathbb{Z}, 1 \cdot 1)$$

as coarsenings.

$$(\mathbb{G}_N]_{\text{sol}_{\mathbb{G}_N}})$$

$$\bar{\Psi}(2)$$

$$\bar{\Psi}(1)$$

$$\bar{\Psi}(0)$$

$$(\mathbb{G}_G]_{\text{sol}_G})$$



bounded

5/1

1 point

112

$$(\mathbb{Z}_{1.1})$$

φ

Can show that this is "the" coarse kernel.

For $[C_1, d_1] \dots [C_m, d_m]$,

$$\mathbb{Z} \ni k$$

$$\mapsto [C_1^k, d_1] \dots [C_m^k, d_m]$$

"pseudo-power map"

3. $\text{view}(\mathcal{B})$: inv qm & bilinear form

Here for $l \geq 2$,

$$G (= \widetilde{\text{Symp}}_0(\Sigma_l, \omega))$$

$$= \text{Symp}_0(\Sigma_l, \omega)$$

$$N (= \widetilde{\text{Ham}}(\Sigma_l, \omega))$$

$$= \text{Ham}(\Sigma_l, \omega)$$

$$\leadsto \Gamma = G/N = H^1(\Sigma_l; \mathbb{R})$$

$$p = \text{Flux} : G \rightarrow \Gamma : \text{the}$$

$\left(\widetilde{\text{Flux}}'' \right)$ flux homom

"Thm 4" $(k^2 M^2, 24)$

$$\forall f, g \in G;$$

$$fg = gf$$

$$\Rightarrow b_I(\text{Flux}(f), \text{Flux}(g)) = 0$$

$$\begin{array}{c} \uparrow \\ b_I : \Gamma \times \Gamma \rightarrow \mathbb{R} \end{array} \quad \begin{array}{c} = H^1(\Sigma_2; \mathbb{R}) \end{array}$$

$$\cdot) ([\alpha], [\beta]) \mapsto \int_{\Sigma_2} \alpha \wedge \beta$$

de Rham

$$\cdot) (\alpha, \beta) \mapsto (\alpha \cup \beta)(\underbrace{[\Sigma_2]}_{\hat{=} H_2}) \in \mathbb{R}$$

H^2

H_2

topological

$b_I =$ algebraic intersection
number on $H^1(\Sigma; \mathbb{R})$

Our proof employs

Py's Calabi gm

$$\mu_p \in \mathbb{Q}(N)^G \quad (106)$$

Fact (1) [BANKAGA]

$$N = [N, N]$$

$$(\text{so, } N = [G, G] \text{ \& } V = W)$$

$$(2) [k^2] [\mu_p] \neq 0 \text{ in } W = V$$

cf. "Cor" (ROUSSEAU 1978)

(M, ω) : open symplectic mfd.

$$G_M := \widehat{\text{Symp}}^c(M, \omega)$$

$$N_M := \widehat{\text{Ham}}^c(M, \omega)$$

$$\Gamma_M := H^{1-c}(M; \mathbb{R})$$

$$1 \rightarrow N_M \rightarrow G_M \xrightarrow{\text{Flux}} \Gamma_M \rightarrow 1$$

Then $\forall \hat{f}, \hat{g} \in G_M$;

$$\hat{f}\hat{g} = \hat{g}\hat{f}$$

$$\Rightarrow b_\omega(\widehat{\text{Flux}}(\hat{f}), \widehat{\text{Flux}}(\hat{g})) = 0$$



$$b_\omega([\alpha], [\beta]) := \int_M \alpha \wedge \beta \wedge \omega^{n-1}$$

(de Rham)

symplectic pairing

True Thm of ROUSSEAU (1978)

(Under the setting of "Cor" above.)

$$\dim M = 2n.$$

Then, $\forall \hat{f}, \hat{g} \in G_M;$

$$\text{Cal}([\hat{f}, \hat{g}])$$

$$= n \times \text{bw}(\text{Flux}(\hat{f}), \text{Flux}(\hat{g}))$$

Here $\text{Cal} : N_M \rightarrow \mathbb{R}$

the Calabi form

Rem $\text{Cal} \in H^1(N_M)^{G_M}.$

But NO non-zero (Calabi) form
for closed case because

$\widetilde{\text{Ham}}(M, \omega)$ is perfect (BANTAGA)

True Thm for "Thm4"

Thm5 ($k^2 M^3$, $25+$)

(Under the setting of "Thm4")

$\forall f, g \in G$;

$\mu_I([f, g])$

$\sim b_I(\text{Flux}(f), \text{Flux}(g))$

$D(\mu_I)$

(☆☆)

Since b_I is bilinear,

by letting $f \mapsto f^k$ ($k \in \mathbb{N}$), can

derive the following limit formula

from (☆☆):

$$b_I(\text{Flux}(f), \text{Flux}(g))$$

$$= \lim_{k \rightarrow \infty} \frac{\mu_I([f^k - g])}{k}$$

(☆☆☆)

(☆☆☆) immediately implies "Thm4"!

Idea

$$\begin{array}{ccc} F : \text{free} & \supseteq & K = [FF] \\ \pi \downarrow & \longleftrightarrow j & \\ G = \text{Symp}_0(\Sigma_g, \omega) & & \end{array}$$

$$\begin{array}{ccc} \pi^* \mu_F \in Q(K)^F & = & H^1(K)^F + j^* Q(F) \\ \uparrow & \uparrow & \text{F: free} \\ \mu_F \in Q(N)^G & & \end{array}$$

$$\leadsto \pi^* \mu_I = \underbrace{h}_{H^1(K)^F} + j^* \underbrace{\phi}_{Q(F)}$$

this h corresponds to the
bilinear form b_I

$$F \times F \rightarrow \mathbb{R}; (\hat{f}, \hat{g}) \mapsto h[\hat{f}, \hat{g}]$$

$$\begin{array}{c} \downarrow \curvearrowright \\ \Gamma \times \Gamma \end{array}$$

this gives the **bilinear form b_I !!**

For a general group G & $N = [G, G]$,

get

$$V(G, N) \hookrightarrow \left\{ \begin{array}{l} \mathbb{R}\text{-valued alternating} \\ \mathbb{Z}\text{-bilinear} \\ \text{form on } \Gamma = G/N \end{array} \right\}$$

by "commutator homogenization"



$$\mu \in \mathbb{Q}(N)^G$$

$\downarrow$

$$G \times G \rightarrow \mathbb{R} : (f, g) \mapsto \mu([f, g])$$

$\uparrow$
decompose this into

$$\left(\begin{array}{c} \text{good} \\ \text{part} \end{array} \right) + \left(\begin{array}{c} \text{bounded} \\ \text{noise} \end{array} \right)$$

$\searrow$ by pseudo-power map

bilinear form.

$$(f, g) \mapsto \lim_{k \rightarrow \infty} \frac{\mu([f^k, g])}{k}$$

SHELOKHIN (14) constructed

$\mu_{\text{Sh}} \in \mathbb{Q}(\widehat{\text{Ham}})^{\text{Symp}_0}$ for every

closed symplectic mfd (M, ω) .

THANK

YOU !!